Cooling Load Chart

A heat load in kilowatts expressed as BTU per hour and tons, and the airflow and chilled water flow needed to carry it away.

One kilowatt of heat is 3,412 BTU per hour, and one ton of cooling is 12,000 BTU per hour, or 3.517 kW. Nearly all the electricity an IT load draws leaves as heat, so a 100 kW room needs about 28.4 tons of sensible cooling before any allowance for other loads or redundancy.

Heat load

kW	BTU/h	Tons
1	3,412	0.284
5	17,061	1.42
10	34,121	2.84
20	68,243	5.69
50	170,607	14.2
100	341,214	28.4
250	853,036	71.1
500	1,706,071	142
1,000	3,412,142	284
2,000	6,824,284	569
5,000	17,060,710	1,422

Sensible heat. For IT equipment, the electrical load in kW is the heat load in kW.

Airflow per kW

Air temperature rise	CFM per kW	CFM for 10 kW	CFM for 100 kW
10°F	316	3,159	31,594
15°F	211	2,106	21,063
20°F	158	1,580	15,797
25°F	126	1,264	12,638
30°F	105	1,053	10,531
35°F	90	903	9,027
40°F	79	790	7,898

Standard air at sea level. A wider temperature rise across the equipment needs less air.

Chilled water flow per ton

Water temperature difference	GPM per ton	GPM for 100 tons	GPM for 500 tons
8°F	3.00	300	1,500
10°F	2.40	240	1,200
12°F	2.00	200	1,000
14°F	1.71	171	857
16°F	1.50	150	750

Water temperature difference	GPM per ton	GPM for 100 tons	GPM for 500 tons
18°F	1.33	133	667
20°F	1.20	120	600

Plain water. A wider difference between supply and return needs less flow.

How the values are calculated

$$\text{BTU/h} = \text{kW} \times 3{,}412$$
$$\text{Tons} = \text{BTU/h} \,/\, 12{,}000$$
$$\text{CFM} = \text{BTU/h} \,/\, (1.08 \times \Delta T)$$
$$\text{GPM} = \text{tons} \times 24 \,/\, \Delta T$$

Every figure in the chart comes from the formulas above. Nothing is copied from a published table, so any cell can be checked by hand.

What this chart does not tell you

These are design-point figures for sensible heat. They leave out lighting, people, heat gain through the building, humidification and the cooling plant's own losses, and they include no redundancy. The air constant of 1.08 applies at sea level; at altitude more air is needed. The water constant applies to plain water, and a glycol mix needs more flow.

This is a reference, not a design. It is here to give you the right order of magnitude and a figure to check against. It does not replace the adopted code or a calculation by a qualified engineer.

From voltfield.org/charts/cooling-load-chart.html, where the notes on reading this chart and a calculator for other cases are. Reviewed November 11, 2026. Free to print, copy and share.